

Homework Submission Policy

- **Late submission:**
 - within 24 hours after its due will incur 20% penalty,
 - after 24 hours and within seven days of its due will incur 50% penalty, and
 - after seven days of its due will not be graded.

Note: One minute late is the same as 23 hours late.

Homework Assignment #2

Textbook Exercises (Based on the 9th edition textbook)

2.1-7. Let a random experiment ...

2.2-10. In the casino game ...

2.3-16. Let X equal the number ...

2.4-18. In grouping testing for ...

2.6-9. A store selling newspapers ...

Due: 2:20pm, April 9, 2026

2.1-7. Let a random experiment be the casting of a pair of fair six-sided dice and let X equal the minimum of the two outcomes.

- (a)** With reasonable assumptions, find the pmf of X .
- (b)** Draw a probability histogram of the pmf of X .
- (c)** Let Y equal the range of the two outcomes (i.e., the absolute value of the difference of the largest and the smallest outcomes). Determine the pmf $g(y)$ of Y for $y = 0, 1, 2, 3, 4, 5$.
- (d)** Draw a probability histogram for $g(y)$.

2.2-10. In the casino game called **high-low**, there are three possible bets. Assume that \$1 is the size of the bet. A pair of fair six-sided dice is rolled and their sum is calculated. If you bet **low**, you win \$1 if the sum of the dice is $\{2, 3, 4, 5, 6\}$. If you bet **high**, you win \$1 if the sum of the dice is $\{8, 9, 10, 11, 12\}$. If you bet on $\{7\}$, you win \$4 if a sum of 7 is rolled. Otherwise, you lose on each of the three bets. In all three cases, your original dollar is returned if you win. Find the expected value of the game to the bettor for each of these three bets.

2.2-11. In the gambling game craps (see Exercise 1.3-13), the player wins \$1 with probability 0.49293 and loses \$1 with probability 0.50707 for each \$1 bet. What is the expected value of the game to the player?

2.3-16. Let X equal the number of flips of a fair coin that are required to observe the same face on consecutive flips.

- (a)** Find the pmf of X . HINT: Draw a tree diagram.
- (b)** Find the moment-generating function of X .
- (c)** Use the mgf to find the values of **(i)** the mean and **(ii)** the variance of X .
- (d)** Find the values of **(i)** $P(X \leq 3)$, **(ii)** $P(X \geq 5)$, and **(iii)** $P(X = 3)$.

2.4-18. In group testing for a certain disease, a blood sample was taken from each of n individuals and part of each sample was placed in a common pool. The latter was then tested. If the result was negative, there was no more testing and all n individuals were declared negative with one test. If, however, the combined result was found positive, all individuals were tested, requiring $n+1$ tests. If $p = 0.05$ is the probability of a person's having the disease and $n = 5$, compute the expected number of tests needed, assuming independence.

2.6-9. A store selling newspapers orders only $n = 4$ of a certain newspaper because the manager does not get many calls for that publication. If the number of requests per day follows a Poisson distribution with mean 3,

- (a)** What is the expected value of the number sold?
- (b)** What is the minimum number that the manager should order so that the chance of having more requests than available newspapers is less than 0.05?